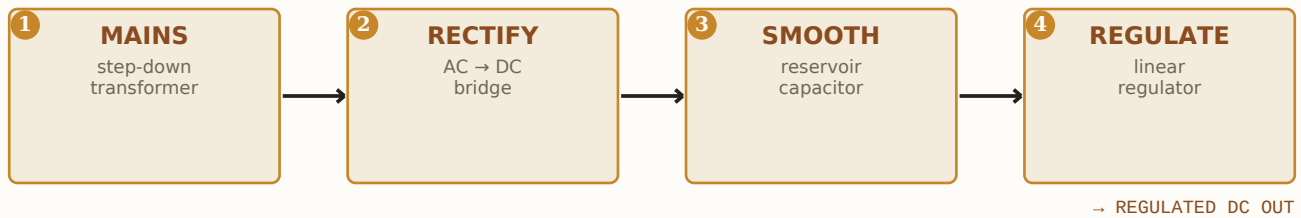


Regulated DC Power Supply

Design, simulation, construction and bench characterisation of a classical linear supply chain.



Overview

This project covered the design, simulation, construction and characterisation of a complete mains-derived regulated DC supply. It implements the classical chain of step-down transformer, rectifier, smoothing stage and linear regulation, and was validated both in simulation and on the bench — giving direct experience of where idealised models and real hardware diverge.

1. Architecture — four power stages

A step-down transformer provides galvanic isolation from the mains and reduces the AC input to a suitable secondary voltage. A rectifier converts the AC waveform to unidirectional current, which is smoothed by a reservoir capacitance sized from the trade-off between ripple amplitude, capacitor stress and rectifier peak current. A linear regulator then holds the output at a fixed DC level, rejecting residual ripple and compensating for load- and input-voltage variation. Each stage demanded quantitative decisions: transformer turns ratio and VA rating, diode peak-inverse-voltage and surge ratings, smoothing capacitance, and the regulator's dropout voltage and dissipation — which directly informed heatsinking.

2. Simulation & verification

The complete supply was modelled in LTSpice before construction. Transient simulation characterised the peak-to-peak ripple at the reservoir capacitor across load conditions and examined the regulator's ripple rejection at the output. Load regulation was assessed by stepping the output current and observing output-voltage deviation, allowing component values to be refined before any hardware was committed.

3. Construction & bench testing

The circuit was built and tested against the simulated predictions. Using an oscilloscope, the ripple waveform at the smoothing stage was measured and compared with the LTSpice results, and the regulated output was verified stable across the intended load range. Working with a mains-derived design also demanded disciplined safety practice — isolation, correct fusing and safe probing technique — reinforcing professional laboratory habits.

Outcomes & skills demonstrated

The finished supply met its design intent, with measured ripple and load-regulation behaviour consistent with simulation. The project demonstrates power-electronics fundamentals, quantitative component sizing, SPICE transient analysis, simulation-to-hardware correlation, safe working practice with mains-derived circuits, and systematic bench characterisation using standard laboratory instruments.

TOOLS

LTSpice · oscilloscope · DMM · electronic load

VALIDATION

sim ↔ bench
correlation